

# Свойства функций Эйри. Точки поворота

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In[268]:=

```
ClearAll["Global`*"]  
$Assumptions =  $\mu > 0$ 
```

Out[269]=

$\mu > 0$

In[270]:=

```
m = 2;
```

## Рекуррентные соотношения

In[271]:=

```
Eq =  $\mu^2 * D[Y[x], \{x, 2\}] - q0[x] * Y[x] == 0$ 
```

Out[271]=

$-q0[x] \times Y[x] + \mu^2 Y''[x] == 0$

In[272]:=

```
sY = Y[x]  $\rightarrow$  a0[x,  $\mu$ ] * v[ $\eta[x, \mu]$ ] +  $\mu^{(1 / 3)}$  * a1[x,  $\mu$ ] * D[v[ $\eta[x, \mu]$ ],  $\eta[x, \mu]$ ]
```

Out[272]=

$Y[x] \rightarrow a0[x, \mu] \times v[\eta[x, \mu]] + \mu^{1/3} a1[x, \mu] v'[\eta[x, \mu]]$

In[273]:=

```
sa0 = a0[x,  $\mu$ ]  $\rightarrow$  Sum[a0[i][x] *  $\mu^i$ , {i, 0, m}]
```

Out[273]=

$a0[x, \mu] \rightarrow a0[0][x] + \mu a0[1][x] + \mu^2 a0[2][x]$

In[274]:=

$$\mathbf{sa1} = \mathbf{a1}[\mathbf{x}, \mu] \rightarrow \text{Sum}[\mathbf{a1}[\mathbf{i}][\mathbf{x}] * \mu^{\mathbf{i}}, \{\mathbf{i}, 0, \mathbf{m}\}]$$

Out[274]=

$$\mathbf{a1}[\mathbf{x}, \mu] \rightarrow \mathbf{a1}[0][\mathbf{x}] + \mu \mathbf{a1}[1][\mathbf{x}] + \mu^2 \mathbf{a1}[2][\mathbf{x}]$$

## Уравнения для определения коэффициентов $a_i^{(j)}(x)$

In[275]:=

$$\mathbf{sn} = \eta[\mathbf{x}, \mu] \rightarrow \mu^{(-2/3)} * \xi[\mathbf{x}]$$

Out[275]=

$$\eta[\mathbf{x}, \mu] \rightarrow \frac{\xi[\mathbf{x}]}{\mu^{2/3}}$$

In[276]:=

$$\mathbf{sv0} = \mathbf{D}[\mathbf{v}[\eta[\mathbf{x}, \mu]], \{\eta[\mathbf{x}, \mu], 2\}] \rightarrow \mathbf{v}[\eta[\mathbf{x}, \mu]] * \eta[\mathbf{x}, \mu]$$

Out[276]=

$$\mathbf{v}''[\eta[\mathbf{x}, \mu]] \rightarrow \mathbf{v}[\eta[\mathbf{x}, \mu]] \times \eta[\mathbf{x}, \mu]$$

In[277]:=

$$\mathbf{sv1} = \mathbf{D}[\mathbf{sv0}, \mathbf{x}] /. \{\mathbf{D}[\eta[\mathbf{x}, \mu], \mathbf{x}] \rightarrow 1\}$$

Out[277]=

$$\mathbf{v}^{(3)}[\eta[\mathbf{x}, \mu]] \rightarrow \mathbf{v}[\eta[\mathbf{x}, \mu]] + \eta[\mathbf{x}, \mu] \mathbf{v}'[\eta[\mathbf{x}, \mu]]$$

In[278]:=

$$\mathbf{sq0} = \mathbf{q0}[\mathbf{x}] \rightarrow \xi[\mathbf{x}] * \mathbf{D}[\xi[\mathbf{x}], \mathbf{x}]^2$$

Out[278]=

$$\mathbf{q0}[\mathbf{x}] \rightarrow \xi[\mathbf{x}] \xi'[\mathbf{x}]^2$$

In[279]:=

```
Eq1 = Eq /. Table[D[sY, {x, i}], {i, 0, 2}]
```

Out[279]=

$$\begin{aligned}
& -q0[x] \left( a0[x, \mu] \times v[\eta[x, \mu]] + \mu^{1/3} a1[x, \mu] v'[\eta[x, \mu]] \right) + \\
& \mu^2 \left( 2 v'[\eta[x, \mu]] a0^{(1,0)}[x, \mu] \eta^{(1,0)}[x, \mu] + 2 \mu^{1/3} v''[\eta[x, \mu]] a1^{(1,0)}[x, \mu] \eta^{(1,0)}[x, \mu] + v[\eta[x, \mu]] a0^{(2,0)}[x, \mu] + \right. \\
& \mu^{1/3} v'[\eta[x, \mu]] a1^{(2,0)}[x, \mu] + a0[x, \mu] \left( v''[\eta[x, \mu]] \eta^{(1,0)}[x, \mu]^2 + v'[\eta[x, \mu]] \eta^{(2,0)}[x, \mu] \right) + \\
& \left. \mu^{1/3} a1[x, \mu] \left( v^{(3)}[\eta[x, \mu]] \eta^{(1,0)}[x, \mu]^2 + v''[\eta[x, \mu]] \eta^{(2,0)}[x, \mu] \right) \right) = 0
\end{aligned}$$

In[280]:=

```
Eq2 = Collect[Eq1 /. {sv0, sv1}, {v[\eta[x, \mu]], D[v[\eta[x, \mu]], \eta[x, \mu]]}]
```

Out[280]=

$$\begin{aligned}
& v'[\eta[x, \mu]] \\
& \left( -\mu^{1/3} a1[x, \mu] \times q0[x] + 2 \mu^2 a0^{(1,0)}[x, \mu] \eta^{(1,0)}[x, \mu] + \mu^{7/3} a1[x, \mu] \times \eta[x, \mu] \eta^{(1,0)}[x, \mu]^2 + \mu^{7/3} a1^{(2,0)}[x, \mu] + \mu^2 a0[x, \mu] \eta^{(2,0)}[x, \mu] \right) + \\
& v[\eta[x, \mu]] \left( -a0[x, \mu] \times q0[x] + 2 \mu^{7/3} \eta[x, \mu] a1^{(1,0)}[x, \mu] \eta^{(1,0)}[x, \mu] + \mu^{7/3} a1[x, \mu] \eta^{(1,0)}[x, \mu]^2 + \right. \\
& \left. \mu^2 a0[x, \mu] \times \eta[x, \mu] \eta^{(1,0)}[x, \mu]^2 + \mu^2 a0^{(2,0)}[x, \mu] + \mu^{7/3} a1[x, \mu] \times \eta[x, \mu] \eta^{(2,0)}[x, \mu] \right) = 0
\end{aligned}$$

In[281]:=

```
Eq3 = Simplify[{Eq2[[1, 1, 2]] == 0, Eq2[[1, 2, 2]] == 0} /. Join[Table[D[s\eta, {x, i}], {i, 0, 2}], {sq0}]]
```

Out[281]=

$$\{a0[x, \mu] \xi''[x] + 2 \xi'[x] a0^{(1,0)}[x, \mu] + \mu a1^{(2,0)}[x, \mu] == 0, a1[x, \mu] (\xi'[x]^2 + \xi[x] \xi''[x]) + 2 \xi[x] \xi'[x] a1^{(1,0)}[x, \mu] + \mu a0^{(2,0)}[x, \mu] == 0\}$$

## Определяем коэффициенты $a_i^{(j)}(x)$

In[282]:=

```
Eq4 = Eq3 /. Flatten[Table[D[{sa0, sa1}, {x, i}], {i, 0, 2}]]
```

Out[282]=

$$\begin{aligned}
& \{2 \xi'[x] (a0[0]'[x] + \mu a0[1]'[x] + \mu^2 a0[2]'[x]) + (a0[0][x] + \mu a0[1][x] + \mu^2 a0[2][x]) \xi''[x] + \mu (a1[0]''[x] + \mu a1[1]''[x] + \mu^2 a1[2]''[x]) == 0, \\
& 2 \xi[x] \xi'[x] (a1[0]'[x] + \mu a1[1]'[x] + \mu^2 a1[2]'[x]) + \\
& (a1[0][x] + \mu a1[1][x] + \mu^2 a1[2][x]) (\xi'[x]^2 + \xi[x] \xi''[x]) + \mu (a0[0]''[x] + \mu a0[1]''[x] + \mu^2 a0[2]''[x]) == 0\}
\end{aligned}$$

$$a_0^{(0)}(x), a_0^{(1)}(x)$$

In[283]:=

```
Eq00 = Coefficient[Eq4[[1, 1]], μ, 0] == 0
```

Out[283]=

$$2 \xi'[x] a_0[0]'[x] + a_0[0][x] \xi''[x] == 0$$

In[284]:=

```
Sol10 = DSolve[Eq00, a0[0][x], x][[1, 1]] /. C[1] -> 1
```

Out[284]=

$$a_0[0][x] \rightarrow \frac{1}{\sqrt{\xi'[x]}}$$

In[285]:=

```
Sol10 = a1[0][x] -> 0
```

Out[285]=

$$a_1[0][x] \rightarrow 0$$

$$a_1^{(0)}(x), a_1^{(1)}(x)$$

In[286]:=

```
Eq11 = Coefficient[Eq4[[2, 1]], μ, 1] == 0
```

Out[286]=

$$a_1[1][x] \xi'[x]^2 + 2 \xi[x] \xi'[x] a_1[1]'[x] + \xi[x] \times a_1[1][x] \xi''[x] + a_0[0]''[x] == 0$$

In[287]:=

```
Sol11 = DSolve[Eq11, a1[1][x], {x, 0, 1}][[1, 1]] /. C[1] -> 0;
Sol11 = Sol11 /. {Sol11[[2, 2, 2]] -> {K[1], c, x}}
```

Out[288]=

$$a1[1][x] \rightarrow \frac{\int_c^x -\frac{a0[0]''[K[1]]}{2\sqrt{\xi[K[1]]}\sqrt{\xi'[K[1]]}} dK[1]}{\sqrt{\xi[x]}\sqrt{\xi'[x]}}$$

In[289]:=

```
Sol01 = a0[1][x] -> 0
```

Out[289]=

```
a0[1][x] -> 0
```

$$a_2^{(0)}(x), a_2^{(1)}(x)$$

In[290]:=

```
Eq02 = Coefficient[Eq4[[1, 1]], \mu, 2] == 0
```

Out[290]=

$$2\xi'[x]a0[2]'[x] + a0[2][x]\xi''[x] + a1[1]''[x] == 0$$

In[291]:=

```
Sol02 = DSolve[Eq02, a0[2][x], {x, 0, 1}][[1, 1]] /. C[1] -> 0;
Sol02 = Sol02 /. {Sol02[[2, 1, 2]] -> {K[1], c, x}}
```

Out[292]=

$$a0[2][x] \rightarrow \frac{\int_c^x -\frac{a1[1]''[K[1]]}{2\sqrt{\xi[K[1]]}} dK[1]}{\sqrt{\xi'[x]}}$$

In[293]:=

**Sol12 = a1[2] [x]  $\rightarrow$  0**

Out[293]=

**a1[2] [x]  $\rightarrow$  0**